impl InverseCDF for CanonicalRV

Yu-Ju Ku, Jordan Awan, Aishwarya Ramasethu, Michael Shoemate

December 2, 2024

Proves soundness of the implementation of `InverseCDF` for `CanonicalRV`.
The implementation computes the inverse CDF of the canonical noise random variable.

1 Hoare Triple

Preconditions

Compiler-verified

- Argument `self` of type `CanonicalRV`.
- Argument `uniform` of type `RBig`.
- Argument `refinements` of type `usize`.
- Generic `R` implements `ODPRound` which denotes the rounding mode, either up or down.

Human-verified

Argument `uniform` is in $[0, 1]$.

Pseudocode

```
class InverseCDF(CanonicalRV):
    # type Edge = RBig

    def inverse_cdf(self, uniform: RBig, _refinements: usize, _R) -> RBig | None:
        f_inv = quantile_cnd(uniform, self.tradeoff, self.fixed_point)  #
        return f_inv * self.scale + self.shift
```

Postcondition

Theorem 1.1. Given a random variable `self` (of type `CanonicalRV`), return `Ok(out)` where `out` is the inverse cumulative distribution function of `CanonicalRV` (which includes a rescale and shift) evaluated at `uniform` with error in direction `R`, or `None`.

The error between `out` and the exactly-computed CDF decreases monotonically as `refinements` increases.

Proof. By the definition of `CanonicalRV`,

- `self.tradeoff` is a symmetric nontrivial tradeoff function
- `self.fixed_point` is the fixed point of `self.tradeoff`, where `tradeoff(fixed_point) = c`.

Therefore the preconditions of `quantile_cnd` are met, so `f_inv` on line 5 is a sample from the canonical noise distribution with shift of zero and scale of one.

The function then returns the outcome, scaled by `self.scale` and shifted by `self.shift`.

F_f^{-1}, rescale and shift are computed exactly via fractional arithmetic, so making use of refinements is not necessary, as the error in computing the inverse CDF is already zero, satisfying the monotonicity property of the error.

□